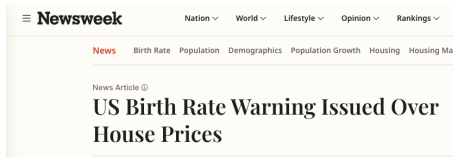


Fertility and Housing Tenure Choice

Tommaso De Santo

September 14, 2026

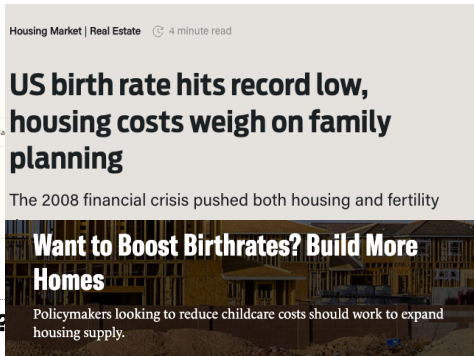
Housing and Fertility



POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.



Renting a home 'leads to lower birth rates'

Housing, Fertility, and Tenure Choice

Housing costs affect fertility, but tenure and wealth matter:

- Earlier access to homeownership increases family size (Fazio et al. 2025)
- Credit access raises buying, births, and home size (Hacamo 2021, Couillard 2025)

Children increase how big of a house you need. Key channels:

- Owning: tenure segmentation (big houses are often only for sale) [▶ Data](#)
- Wealth: borrowing constraints affect purchase ability
- Intergenerational distribution: old people own most of the large dwellings [▶ Data](#)

How much can physical space affect fertility?

What kind of housing policy is good for natality?

This Paper

Question: how much does access to space shape fertility, and can housing policy change it?

Model: a lifecycle economy with one housing market, in which children need space

- Households choose when to try for children; children raise the space they need.
- Renting is capped in size: large homes must be bought (tenure segmentation).
- Buying requires a down payment: wealth and its timing matter (borrowing constraints).
- Households age, save, and bequeath: large homes end up with the old (intergenerational allocation).

Quantification: the U.S. as a transition to a smaller population

- Calibrate the 2007 economy; fit the 2007–2023 fertility decline with preference shocks.
- Treat 2023 as a point on the transition and project forward.
- Policy: can a property tax that shifts space to young families raise births?

Model

Environment

- Discrete time, overlapping generations of households
- Adult age $a \in [0, J]$ with survival probability s_a ; working age $a \in [0, a_R]$; fertile window $a \in [a_F^0, a_F^1]$
- Children:
 - mature stochastically; housing need depends on children at home
 - are dependents and do not work
- Households differ in wealth, housing, family composition, and idiosyncratic labor efficiency z_t
- Households choose:
 - **Fertility**: whether to attempt the next birth, subject to EV shocks
 - **Tenure**: rent or buy, with tenure segmentation, subject to EV shocks
 - **Savings**: next-period liquid wealth b_{t+1} ; mortgages secured by the house

Preferences

n_t : children ever born; m_t : children currently at home.

Flow utility. CRRA over a consumption aggregator $\mathcal{C}$ and children at home:

$$u_t(c_t, s_t; m_t) = \frac{\mathcal{C}(c_t, s_t; m_t)^{1-\sigma}}{1-\sigma} + \psi_t m_t, \quad \sigma > 0.$$

- $\mathcal{C}_c, \mathcal{C}_s > 0$, concave; children raise consumption and housing needs: $\mathcal{C}_m < 0$.
- Children raise the marginal value of space: $\mathcal{C}_{sm} > 0$.
- $\psi_t m_t$: direct benefit of children at home; starting a family carries a one-time utility cost ξ .

Housing services. Households either rent or own; χ scales owner services:

$$s_t = \mathbf{1}\{\text{own}\} \chi h_t + \mathbf{1}\{\text{rent}\} h_t^R, \quad \chi > 1.$$

Bequests. Terminal utility over bequeathable wealth w_{t+1}^e (liquid wealth plus housing value):

$$B(w_{t+1}^e) = \theta_0 \frac{(\theta_1 + w_{t+1}^e)^{1-\sigma}}{1-\sigma}.$$

Income and saving.

- After-tax earnings $y_t^d(a, z_t)$ and a lump-sum transfer T_t .
- One-period bond b_{t+1} at gross rate R_b ; owners borrow only against the house.

Owning. Buy $h_t \in [0, \bar{h}]$ at price P_t , arriving with h_{t-1} .

- The house depreciates at rate δ_H and is taxed at rate τ_t^P .
- Selling costs a fraction ψ^s of the sale value.
- Buying requires a down payment; the rest is a mortgage secured by the house:

$$b_t + (1 - \psi^s)P_t h_{t-1} \geq \mathbf{1}\{h_t \neq h_{t-1}\}(1 - \phi)P_t h_t, \quad b_{t+1} \geq -\phi P_t h_t.$$

Renting. Rent $h_t^R \in [0, \bar{h}^R]$ at rent r_t per unit, with $\bar{h}^R < \bar{h}$.

- The largest homes are reachable only through ownership.

Fertility and Child Aging

- During fertile ages, households choose each period whether to attempt a birth.
- The choice is discrete, with an extreme-value taste shock ε_t^F of scale κ_1 for the first birth and κ_C for later births.
- An attempt succeeds with age-specific probability π_a and adds one child.
- Children mature stochastically: each dependent child leaves the household with a fixed probability each period, independently of the others.

Firms and Earnings

- Competitive firms produce with effective labor:

$$Y_t = AL_t, \quad w = A, \quad L_t = \int_{a < a_R} e_a z_t dG_t.$$

- Age- and productivity-dependent earnings:

$$y_t^d(a, z_t) = \begin{cases} (1 - \tau^w) w e_a z_t, & a < a_R, \\ \varpi_t, & a \geq a_R. \end{cases}$$

- e_a : age-earnings profile; z_t : idiosyncratic labor efficiency; G_t : distribution of households over states.
- Retirees receive a pay-as-you-go pension ϖ_t ; payroll taxes balance it each period.

Housing Supply and Rents

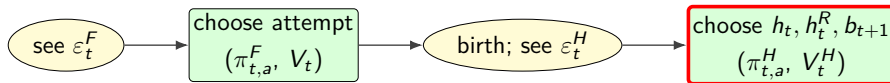
Housing supply:

$$H_t^s = H_0 P_t^\eta.$$

- H_0 : supply scale; η : supply elasticity.
- Rent is the user cost of housing: a competitive landlord who buys at P_t , pays depreciation and property tax, and resells at P_{t+1} must earn the bond return,

$$r_t = (R_b + \delta_H + \tau_t^P)P_t - P_{t+1}.$$

The Household's Problem: Housing and Savings



Conditional on the realized family size, choose housing and savings:

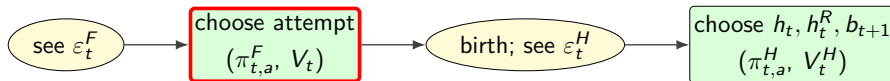
$$V_t^H(b_t, h_{t-1}, z_t, a, n, m) = \mathbb{E}_{\varepsilon_t^H} \max_{h_t, h_t^R, b_{t+1}} \left\{ u_t(c_t, \chi h_t + h_t^R; m) + \varepsilon_t^H(h_t) \right. \\ \left. + \beta \mathbb{E}_t[s_a V_{t+1}(b_{t+1}, h_t, z_{t+1}, a+1, n_{t+1}, m_{t+1}) + (1-s_a)B(w_{t+1}^e)] \right\}.$$

subject to

$$c_t + b_{t+1} + r_t h_t^R + (\delta_H + \tau_t^p) P_t h_t = R_b b_t + y_t^d(a, z_t) + T_t - R_b \mathbf{1}\{h_t \neq h_{t-1}\} P_t [h_t - (1 - \psi^s) h_{t-1}],$$

$$b_{t+1} \geq -\phi P_t h_t, \quad b_t + (1 - \psi^s) P_t h_{t-1} \geq \mathbf{1}\{h_t \neq h_{t-1}\} (1 - \phi) P_t h_t.$$

The Household's Problem: Fertility Choice



Waiting and attempting have values

$$W_t^0 = V_t^H(b_t, h_{t-1}, z_t, a, n_t, m_t),$$

$$W_t^A = \pi_a [V_t^H(b_t, h_{t-1}, z_t, a, n_t + 1, m_t + 1) - \xi \mathbf{1}\{n_t = 0\}] \\ + (1 - \pi_a) V_t^H(b_t, h_{t-1}, z_t, a, n_t, m_t).$$

Before fertility tastes, household value is

$$V_t(b_t, h_{t-1}, z_t, a, n_t, m_t) = \mathbb{E}_{\varepsilon_t^F} \max\{W_t^0 + \varepsilon_{t,0}^F, W_t^A + \varepsilon_{t,A}^F\}.$$

Choice probabilities:

$$\pi_{t,a}^F(b_t, h_{t-1}, z_t, n_t, m_t) = \frac{\exp\{W_t^A/\kappa(n_t)\}}{\exp\{W_t^A/\kappa(n_t)\} + \exp\{W_t^0/\kappa(n_t)\}},$$

Here $\kappa(n_t) = \kappa_1$ if $n_t = 0$ and κ_C otherwise. Expected births equal $\pi_a \pi_{t,a}^F$.

Equilibrium

Given parameters, an initial population N_0 , and an initial household distribution G_0 , an equilibrium is a sequence

$$\{V_t, g_t, G_t, N_t, P_t, r_t, T_t, \varpi_t\}_{t \geq 0}$$

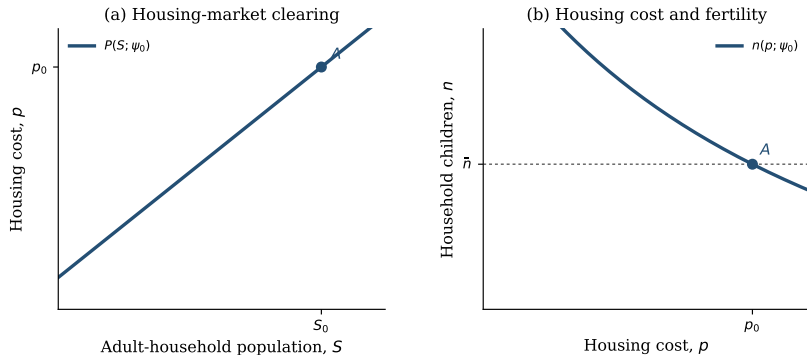
such that, for every t :

1. **Households.** Given $\{P_s, r_s, T_s, \varpi_s\}_{s \geq t}$, V_t solves the household problem with continuation V_{t+1} ; the policy functions g_t (fertility, housing, savings) attain it.
2. **Markets and government.** Rent is the user cost of housing, the housing market clears, pensions balance payroll taxes, and property-tax revenue is rebated equally.
3. **Population.** Births $B_t = \int \pi_a \pi_{t,a}^F dG_t$ enter at age zero and persons age with survival s_a : $N_{0,t+1} = B_t$, $N_{a+1,t+1} = s_a N_{a,t}$.
4. **Consistency.** G_{t+1} follows from G_t under g_t and the idiosyncratic shocks, with entrants making the number of households at each age consistent with N_{t+1} .

Quantification

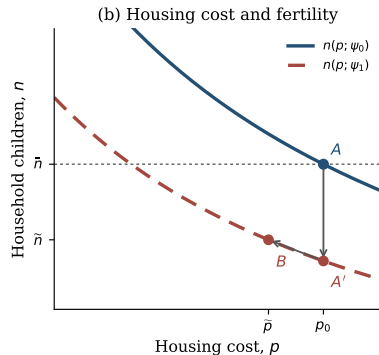
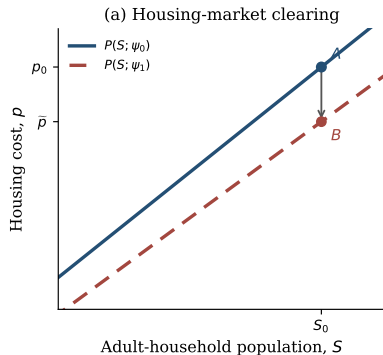
Initial Steady State

For intuition, summarize housing costs by p and fertility by $n(p; \psi)$.



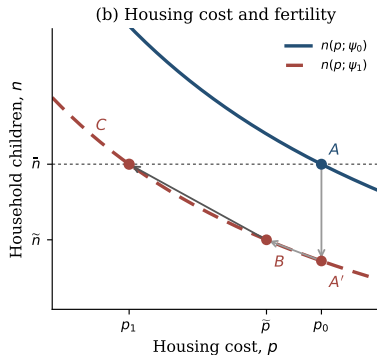
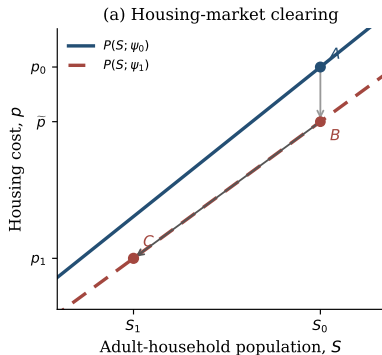
At A , housing clears at (S_0, p_0) and fertility is at the replacement rate $\bar{n}$.

Impact



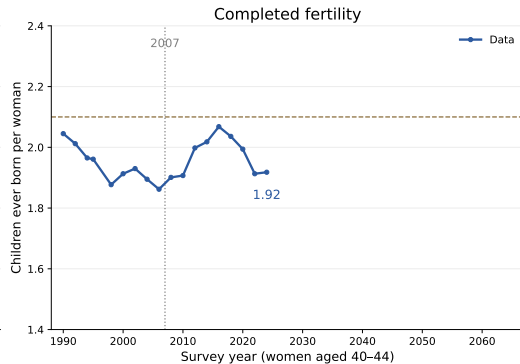
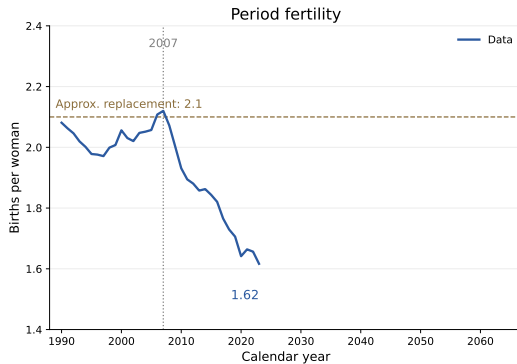
A fall from ψ_0 to ψ_1 lowers fertility at unchanged housing cost: $A \rightarrow A'$. Housing costs then adjust at inherited population S_0 : $A' \rightarrow B$.

Demographic Adjustment

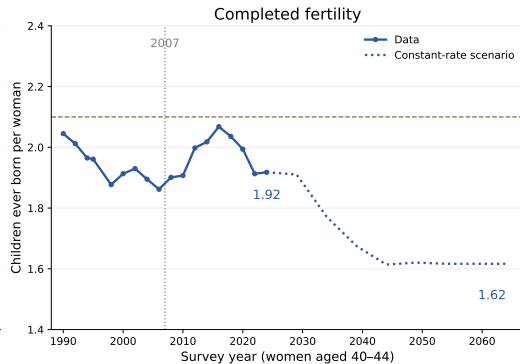
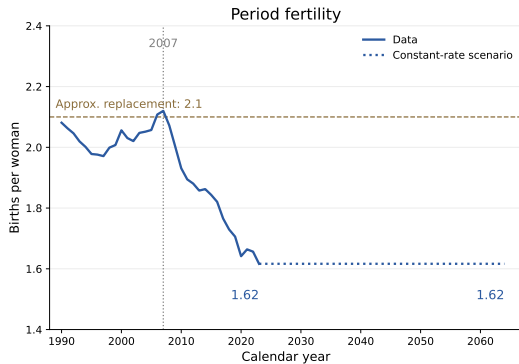


Smaller entering cohorts reduce housing demand. Lower housing costs partly offset the preference decline; C marks the new stationary outcome.

Fertility in the United States



Fertility in the United States



1. Calibrate the 2007 steady state.

Match fertility, housing, and wealth moments to determine household parameters.

2. Estimate the sequence of fertility shocks.

Hold those parameters fixed and match birth rates through 2023 with unanticipated changes in fertility preferences. Carry households and population forward between shocks.

3. Project the economy forward.

After 2023, hold fertility preferences fixed and solve for prices, housing choices, and population. Compare housing policies from the same 2023 economy.

A model period is four calendar years.

Utility Specification

$$u_t(c_t, s_t; m_t) = \frac{\left[\frac{c_t^{\alpha_0} (s_t - \bar{h}(m_t))^{1-\alpha_0}}{e(m_t)} \right]^{1-\sigma}}{1-\sigma} + \psi_t m_t.$$

$$e(m_t) = \left(\frac{2 + \omega m_t}{2} \right)^\nu, \quad \bar{h}(m_t) = h_P \mathbf{1}\{m_t > 0\}.$$

Equivalence scale $e(m_t)$ as in Scholz, Seshadri & Khitatrakun (2006).

Identification

Ten parameters are chosen jointly to match twelve moments.

Fertility

- ψ_0 , preference for children $\rightarrow$ completed fertility
- ξ , first-birth cost $\rightarrow$ childlessness
- κ_1 , first-birth taste dispersion $\rightarrow$ mean age at first birth, first births after 30
- κ_C , later-birth taste dispersion $\rightarrow$ one-child families

Housing

- h_P , space requirement of parents $\rightarrow$ rooms added at first birth, rooms of 3+ versus 1–2 children
- χ , owner premium $\rightarrow$ ownership rate
- H_0 , supply scale $\rightarrow$ mean rooms

Saving and bequests

- β , discount factor $\rightarrow$ wealth over earnings
- θ_0 , bequest scale $\rightarrow$ bequest flow
- θ_1 , bequest wealth shift $\rightarrow$ old-age p90/median

Calibration: Targets and Model

Moment	Target	Model
Completed fertility	2.10	2.10
Childless, women 40–44 (%)	19.8	20.7
Exactly one child among mothers, 40–44 (%)	21.4	20.7
Mean age at first birth (years)	25.98	26.10
First births at age 30+ (%)	24.9	23.2
Mean occupied rooms (capped at 9)	5.56	6.29
Ownership, heads 30–55 (%)	64.8	52.4
First-birth room response, –1 to +3	0.720	1.011
Rooms: 3+ versus 1–2 resident children	0.347	0.200
Wealth / annual gross earnings	6.15	4.85
Annual bequests / wealth (%)	0.88	0.86
Wealth/income p90 / median, ages 76–84	3.52	4.44

Sources: CPS 2004/2006 (children ever born, women 40–44); natality records 2003–2006 (first-birth timing); ACS 2005/2006, 42 metros (rooms, ownership); PSID 2003/2005 (wealth); PSID event study (room response).

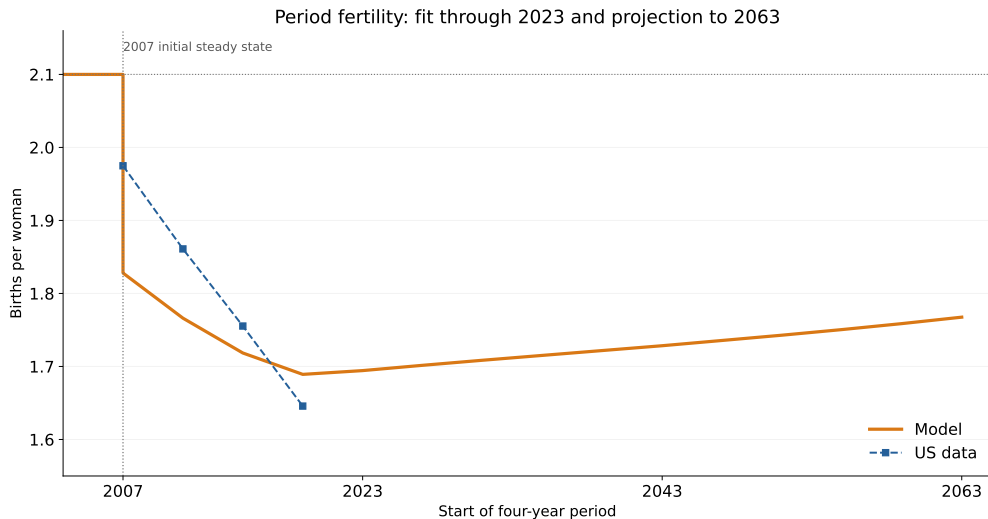
Internally Calibrated Parameters

Parameter	Economic interpretation	Value
β_{annual}	Annual discount factor	0.990
κ_1	First-birth taste dispersion	0.279
κ_C	Later-birth taste dispersion	0.347
χ	Owner housing-service multiplier	1.048
H_0	Housing supply scale	8.191
θ_0	Bequest-motive scale	0.087
θ_1	Bequest-utility wealth shift	0.057
ξ	First-birth utility cost	0.127
h_P	Parenthood space requirement	2.300
ψ_0	Initial preference for children	0.160

Externally Calibrated Parameters

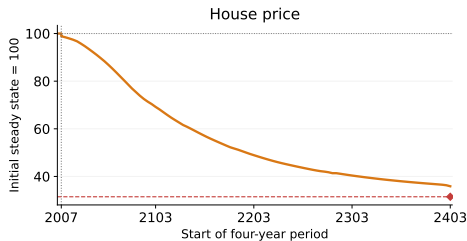
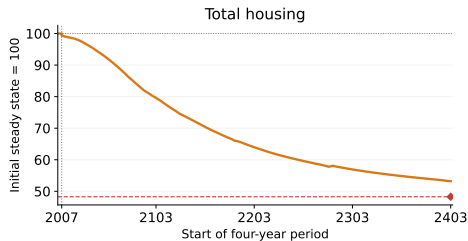
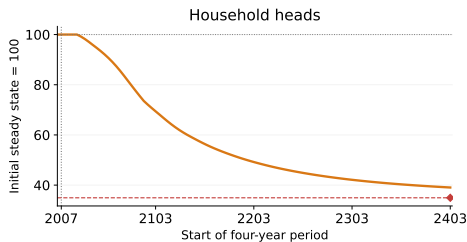
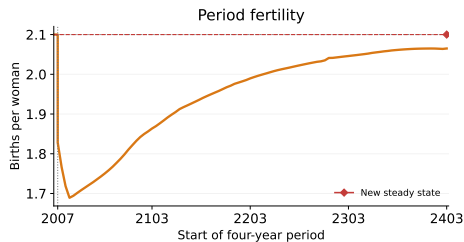
Parameter	Value	Source
Risk aversion σ	2	
Equivalence scale (ω, ν)	(0.7, 0.7)	Scholz et al. (2006)
Consumption share α_0	0.733	CEX 2019–23, childless renters
Housing-supply elasticity η	0.63	Baum-Snow & Han (2024)
Annual real interest rate q_{ann}	2.0%	Greaney et al. (2025)
Payroll tax rate τ^w	17.9%	OECD U.S. average
Financed share ϕ	0.80	20% down payment
Depreciation δ_H , property tax τ^p	2%, 1% per year	
Selling cost ψ^s	6%	Greaney et al. (2025)
Rental size cap $\bar{h}^R$	6 rooms	Greaney et al. (2025)
Ages	18, 66, 45	entry, retirement, last fertile age

Fertility Transition: 2007–2063



Full Transition: 2007–2403

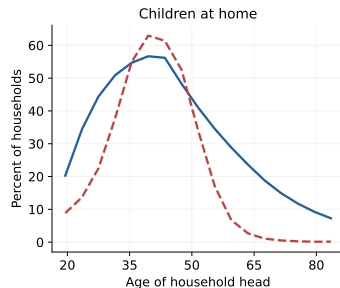
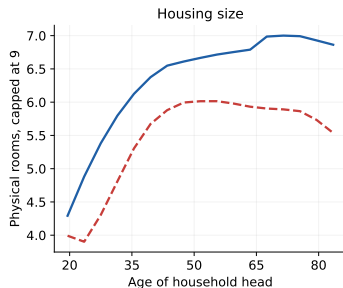
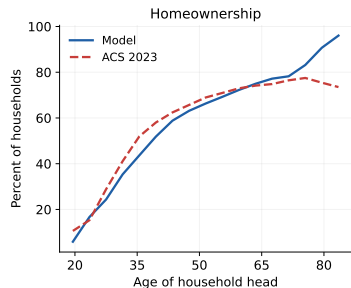
Full transition after a permanent fertility-preference decline



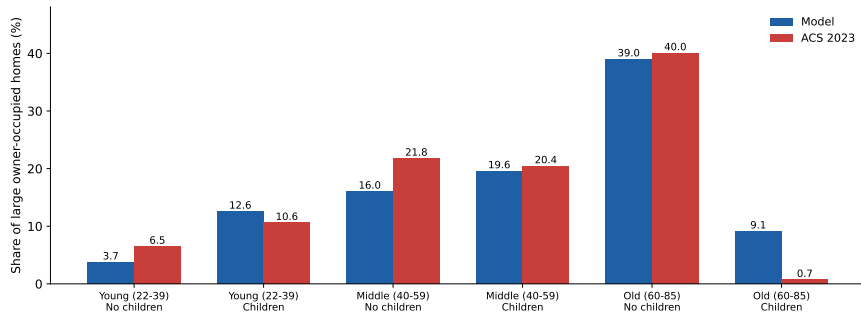
2023 Data and Model

Moment	Data	Model
Completed fertility, ages 40–44	1.92	1.70
Childless, ages 40–44 (%)	18.8	25.0
Exactly one child among mothers, 40–44 (%)	22.2	25.5
Mean age at first birth (years)	28.09	29.74
First births at age 30+ (%)	37.7	44.0
Mean occupied rooms (capped at 9)	5.58	6.07
Ownership, heads 30–55 (%)	58.7	46.9
First-birth room response, –1 to +3	0.720	1.060
Rooms: 3+ versus 1–2 resident children	0.355	0.266
Wealth / annual gross earnings	6.87	5.64
Annual bequests / wealth (%)	0.88	0.70
Wealth/income p90 / median, ages 76–84	3.45	3.50

Lifecycle Fit in 2023



Intergenerational Allocation: Model vs Data



Share of owner-occupied homes with 6+ rooms; ACS 2023.

- Large homes sit with older households without children; young families hold few.
- Coven et al. (2025): a rebated property tax moves space from old owners to young households.
- Those young households are the ones deciding on children, and children need space.

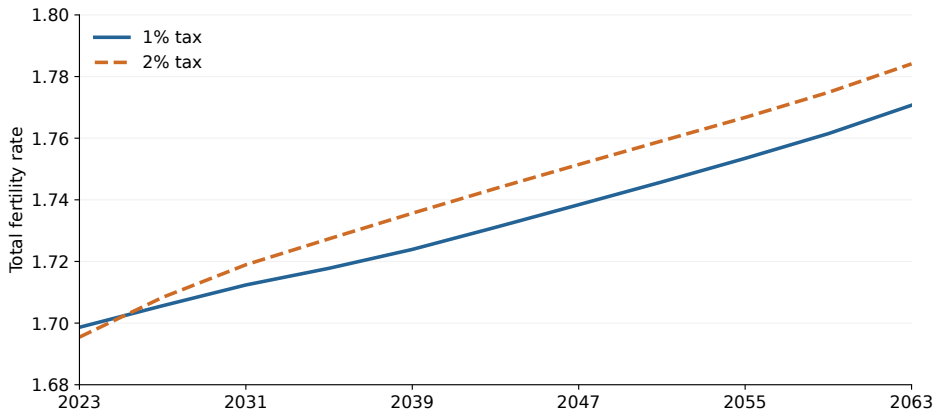
The experiment of Coven, Golder, Gupta & Ndiaye (2025)

- Tax housing by value; rebate the revenue equally to all households.
- Older owners pay more than they get back; young renters get back more than they pay.
- House prices fall; constrained young households buy sooner or buy bigger.

This paper

- Same experiment: property tax from 1% to 2%, equal rebate, from the 2023 economy.
- Follow prices, rents, who lives in the large homes, and births along the path.
- Does the space that moves to young families turn into children?

Policy Results



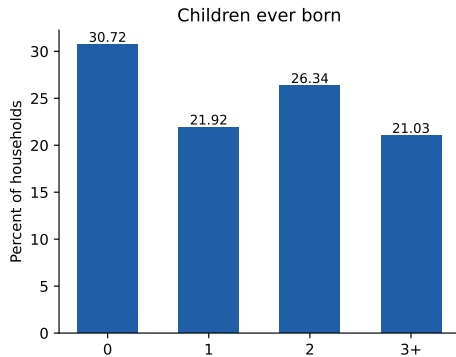
Conclusion

- A lifecycle model of fertility with realistic housing: tenure, down payments, space
- Children need space; large homes must be bought; the old hold most of them
- Today is a point on a transition to a smaller population
- A rebated property tax can move space toward young families, generating a fertility response

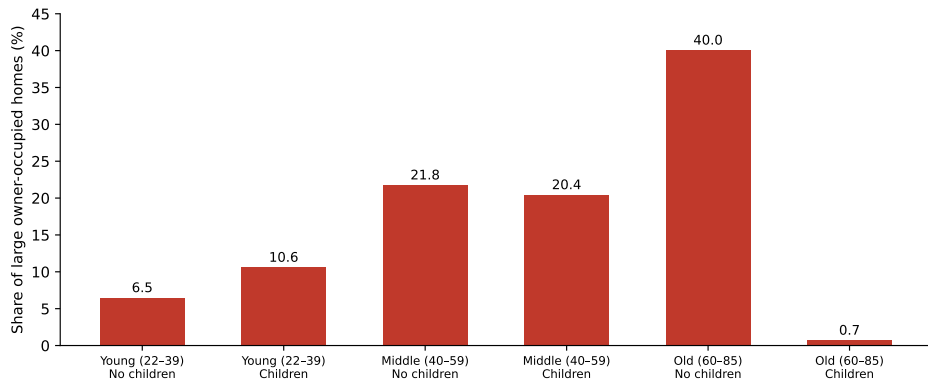
Thank You!

Appendix

The 2023 Equilibrium

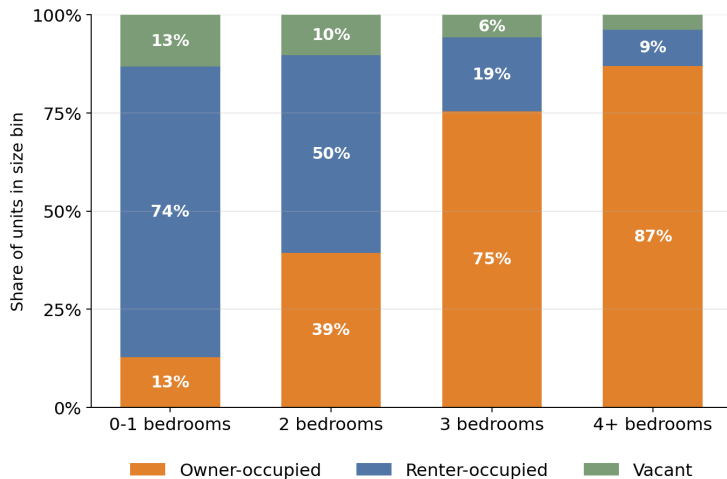


Who Owns the Large Homes



Owner-occupied homes with 6+ rooms; ACS 2023.

Big Units Are Owned

[▶ Back](#)

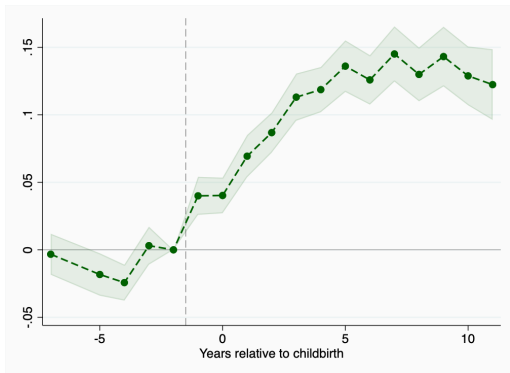
Source: American Housing Survey, 2023.

- **AHS and ACS:** housing sizes, tenure, household composition, and age profiles.
- **PSID:** housing around childbirth, earnings, and wealth by age.
- **CPS and natality records:** completed fertility, childlessness, and birth timing.
- **Census population estimates:** age–sex population and demographic flows.

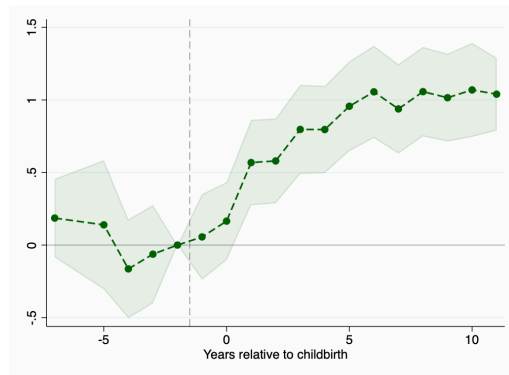
Cross-sectional housing patterns discipline access to space; longitudinal data describe housing adjustment around births.

Ownership and Housing Size Around First Birth

Ownership probability



Number of rooms

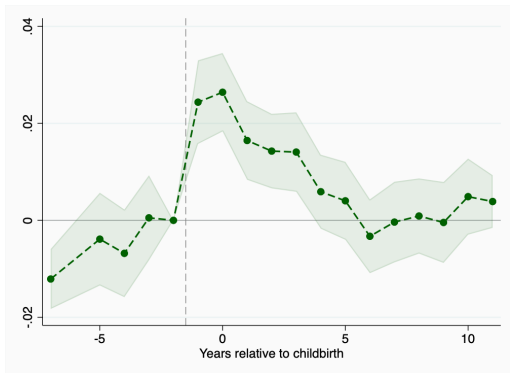


Original May event-study coefficients; measurement revisions under review.

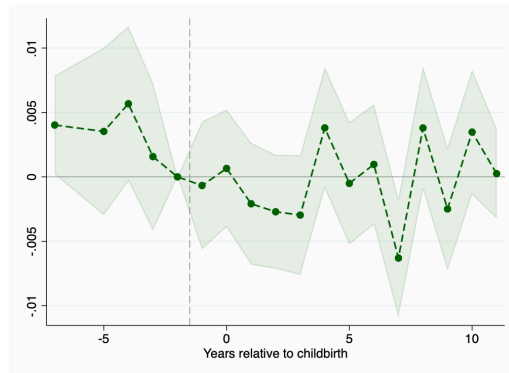
Source: PSID; original May event-study estimates.

Moving Around First Birth

Expansion or better housing



Neighborhood, schools, or proximity



Original May event-study coefficients; measurement revisions under review.

Source: PSID; original May event-study estimates.

Steady-State Solution

Solve for the house price P , pension ϖ , and rebate T , given parameters and a fixed fertility preference.

1. Guess (P, ϖ, T) ; derive rent r from the user cost.
2. Solve households backward over age; compute choice probabilities.
3. Construct the stationary household distribution. Set household mass equal to housing supply divided by mean housing per household.
4. Compute three residuals: housing excess demand, the pension budget, and the rebate budget. The loss is the largest residual in absolute value.
5. Update (P, ϖ, T) and repeat until the loss is below tolerance. Then apply the population law for one period and verify that the distribution reproduces itself.

Transition Solution

Solve for the paths $(P_t, \varpi_t, T_t)_{t \geq t_0}$, given inherited households and a preference believed permanent.

1. Solve the new steady state; it supplies the terminal value function.
2. Guess the three paths; derive rents from the user cost.
3. Solve households backward in calendar time from the terminal value.
4. Advance households and entry queues forward from the inherited state; aggregate housing demand and taxes at each date.
5. Compute the three residuals at every date. The loss is the largest residual across dates.
6. Update the paths jointly and repeat until the loss is below tolerance. Extend the horizon to check that early responses are stable.